

Neuromorphic Radar Signal Processing for Stealth Aircraft Detection: Bio-Inspired Visual Cortex Algorithms Applied to Low-Observable Target Extraction in Heavy Clutter Environments

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Abstract

Modern stealth aircraft achieve radar cross sections (RCS) below 0.001 m^2 , rendering them nearly invisible to conventional constant false alarm rate (CFAR) detectors that rely on amplitude thresholding alone. This paper introduces a neuromorphic radar signal processing pipeline that translates known biological visual cortex mechanisms—lateral inhibition, oriented Gabor filtering, temporal integration, and predictive coding—into the radar domain. In our computational framework, a simulated pulse-Doppler radar with bandwidth $B = 10 \text{ MHz}$ observes a Swerling-I target embedded in Weibull-distributed ground clutter (shape parameter $c = 1.6$) and Gaussian thermal noise. The neuromorphic pipeline replaces the standard cell-averaging CFAR with a four-stage bio-inspired processing chain: (i) centre-surround lateral inhibition that suppresses spatially correlated clutter, reducing clutter variance by a factor of 3.8; (ii) a bank of Gabor filters tuned to 12 orientations that extract coherent target signatures from the range-Doppler map; (iii) a leaky-integrate-and-fire (LIF) temporal accumulator that sums weak detections across 32 coherent processing intervals (CPI), improving effective signal-to-noise ratio by roughly 11.2 dB; and (iv) a predictive-coding anomaly detector that flags

deviations from a first-order autoregressive clutter model. In Monte Carlo trials (10,000 runs), the neuromorphic pipeline achieves a detection probability $P_d = 0.87$ for a -20 dBsm target at $\text{SNR} = -8$ dB, compared with $P_d = 0.23$ for CA-CFAR under identical conditions—a 3.8-fold improvement. The false alarm rate remains controlled at $P_{fa} = 10^{-4}$. These preliminary results suggest that neuroscience-inspired architectures may offer a complementary pathway for low-observable target detection.

Keywords: neuromorphic radar; stealth aircraft detection; bio-inspired signal processing; visual cortex; lateral inhibition; Gabor filtering; CFAR; radar cross section; low-observable targets; predictive coding

1 Introduction

The evolution of stealth technology has fundamentally altered the calculus of air defence. Fifth-generation fighters such as the F-22 Raptor, F-35 Lightning II, and Sukhoi Su-57 achieve frontal radar cross sections on the order of $\sigma \approx 0.0001\text{--}0.001\text{ m}^2$ [1]. At these levels, a conventional ground-based radar operating at S-band (2–4 GHz) must compensate by increasing transmit power by a factor of roughly 10^5 relative to a conventional fighter target ($\sigma \approx 5\text{ m}^2$)—a requirement that is both economically prohibitive and tactically impractical for mobile air defence units. The alternative, which this paper explores, is to process the existing received signal far more intelligently. Our confidence in the specific quantitative claims remains somewhat tempered by the simulation-only nature of these experiments.

The standard detection paradigm in modern pulse-Doppler radars follows a well-established chain: matched filtering, Doppler processing via the fast Fourier transform (FFT), and constant false alarm rate (CFAR) detection [2]. While this pipeline is highly effective against conventional targets, it suffers from three well-documented limitations when confronted with low-observable (LO) threats. First, CFAR detectors assume that the clutter statistics in the reference cells surrounding the cell under test (CUT) are stationary and homogeneous; this assumption breaks down in heterogeneous environments such as littoral zones, mountainous terrain, and urban settings [3]. Second, amplitude-only thresholding discards phase and Doppler texture information that may distinguish a slowly moving stealth target from clutter edges. Third, single-CPI detection provides no mechanism for accumulating weak returns across multiple looks at the target.

Intriguingly, the mammalian visual system faces an analogous challenge every time a camouflaged animal must be detected against a textured, noisy background. Decades of neuroscience research have elucidated the cortical mechanisms that solve this problem with remarkable efficiency [4]. In the primary visual cortex (V1), lateral inhibition enhances local contrast by suppressing responses from neighbouring neurons with similar spatial

frequencies [5]. Simple cells in V1 act as oriented bandpass filters well described by Gabor functions [6]. Temporal integration in higher areas (V4, MT) accumulates weak signals across multiple time frames. Predictive coding, implemented through top-down feedback connections, highlights stimuli that deviate from the brain’s internal model of the environment [7]. These findings raise a natural question: could translating these biological signal processing strategies into radar domain yield detection performance improvements against low-RCS targets, without requiring any hardware modifications?

Several groups have explored bio-inspired approaches to radar in recent years. Haykin proposed cognitive radar, in which a feedback loop between the transmitter and receiver adaptively adjusts waveform parameters based on the perceived environment—an architecture loosely inspired by the perception-action cycle in neuroscience [8]. Zhang and colleagues implemented a spiking neural network for ISAR image classification, demonstrating energy-efficient processing but not addressing detection per se [9]. Martone and colleagues at the US Army Research Laboratory used reinforcement learning for cognitive electronic warfare, where the radar “learns” to counter jamming—again tangentially bio-inspired but not derived from specific cortical mechanisms [10]. To the best of our knowledge, no prior study has systematically translated the four-stage V1 processing hierarchy (lateral inhibition → oriented filtering → temporal integration → predictive coding) into a coherent radar detection pipeline and benchmarked it against CFAR in Monte Carlo simulation.

The contributions of this paper are as follows. First, we formally define the neuromorphic radar processing pipeline (Section 3) and derive closed-form expressions for the clutter suppression gain of each stage. Second, we present a full Monte Carlo evaluation (Section 4) comparing the neuromorphic pipeline with cell-averaging CFAR (CA-CFAR), ordered-statistic CFAR (OS-CFAR), and a deep-learning baseline (convolutional neural network on range-Doppler maps). Third, we provide a computational cost analysis showing that the neuromorphic pipeline operates within the real-time budget of modern radar processors. We emphasise that all results reported here are simulation-based and should be treated as preliminary evidence rather than definitive proof of operational superiority.

2 Background and Related Work

2.1 Radar Detection Fundamentals

The radar range equation governs the received power from a target at range R :

$$P_r = \frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 R^4 L} \quad (1)$$

where P_t is the transmit power, G is the antenna gain, λ is the wavelength, σ is the radar cross section, and L accounts for system losses. The signal-to-noise ratio (SNR) at the receiver output is $\text{SNR} = P_r / (k_B T_s B)$, where k_B is Boltzmann's constant, T_s is the system noise temperature, and B is the receiver bandwidth.

For a Swerling-I target (constant RCS within a single CPI, independent between CPIs), the single-pulse detection probability under Neyman–Pearson criterion is given by the Marcum Q-function:

$$P_d = Q_1\left(\sqrt{2 \text{SNR}}, \sqrt{-2 \ln P_{fa}}\right) \quad (2)$$

At $\text{SNR} = -8$ dB and $P_{fa} = 10^{-4}$, the theoretical single-pulse detection probability for a Swerling-I target is only $P_d \approx 0.03$ —effectively no detection. This sets the baseline against which multi-CPI integration and our neuromorphic processing are compared, though we acknowledge that more sophisticated conventional approaches also exist.

2.2 CFAR Detection

Cell-averaging CFAR (CA-CFAR) estimates the local noise power from N reference cells surrounding the CUT:

$$\hat{\mu} = \frac{1}{N} \sum_{i=1}^N x_i, \quad T = \alpha \hat{\mu} \quad (3)$$

where $\alpha = N (P_{fa}^{-1/N} - 1)$ is the threshold multiplier for exponentially distributed clutter. CA-CFAR is optimal under homogeneous Rayleigh clutter but degrades markedly under clutter-edge (masking) and multi-target (capture) scenarios [11]. OS-CFAR replaces the mean with the k -th order statistic, providing robustness to interfering targets but at the cost of a detection loss of approximately 1–2 dB. Neither approach exploits spatial, spectral, or temporal texture of the radar returns—information that is abundant but discarded by conventional processing.

2.3 Biological Visual Processing

The primate visual cortex processes retinal input through a hierarchical cascade of increasingly abstract representations [4]. At the earliest stage, retinal ganglion cells and lateral geniculate nucleus (LGN) neurons exhibit centre-surround receptive fields that can be modelled as a difference of Gaussians (DoG):

$$\text{DoG}(x, y) = \frac{1}{2\pi\sigma_c^2} \exp\left(-\frac{x^2 + y^2}{2\sigma_c^2}\right) - \frac{1}{2\pi\sigma_s^2} \exp\left(-\frac{x^2 + y^2}{2\sigma_s^2}\right) \quad (4)$$

where σ_c and σ_s are the centre and surround Gaussian widths, respectively, with a typical ratio $\sigma_s/\sigma_c \approx 1.6$. This operation implements spatial high-pass filtering, strongly suppressing smooth, uniform backgrounds while preserving local discontinuities—precisely

the property needed to extract a point target from textured clutter.

In V1, simple cells are selective for orientation and spatial frequency. Their receptive fields are well described by 2D Gabor functions:

$$G(x, y; \theta, f_0, \sigma_g) = \exp\left(-\frac{x'^2 + \gamma^2 y'^2}{2\sigma_g^2}\right) \cos(2\pi f_0 x' + \phi) \quad (5)$$

where $x' = x \cos \theta + y \sin \theta$, $y' = -x \sin \theta + y \cos \theta$, θ is the preferred orientation, f_0 is the spatial frequency, γ is the aspect ratio, and ϕ is the phase [6]. A bank of Gabor filters covering 12 orientations at multiple scales provides a near-complete tiling of the spatial frequency domain.

3 Methods: Neuromorphic Radar Pipeline

3.1 Simulation Environment

We simulate a ground-based pulse-Doppler surveillance radar operating at S-band ($f_c = 3.0$ GHz, $\lambda = 0.1$ m) with bandwidth $B = 10$ MHz (range resolution $\Delta R = 15$ m), pulse repetition frequency $\text{PRF} = 2,000$ Hz, and coherent processing interval $T_{\text{CPI}} = 16$ ms (32 pulses per CPI). The maximum unambiguous range is 180 km and the maximum unambiguous velocity is ± 50 m/s.

The target is modelled as a Swerling-I point scatterer with RCS of -20 dBsm ($\sigma = 0.01$ m², similar to a low-observable aircraft at certain aspect angles) at a range of 120 km. Ground clutter is generated as Weibull-distributed returns with shape parameter $c = 1.6$ and scale parameter adjusted to achieve a clutter-to-noise ratio (CNR) of 35 dB in the mainlobe clutter region—representative of a heterogeneous land background with discrete clutter sources. Thermal noise is white Gaussian with power normalised to 0 dB.

The range-Doppler map (RDM) is formed by pulse compression (matched filtering) along the fast-time axis followed by a 32-point FFT along the slow-time axis. Each pixel of the resulting 1200×32 RDM contains a complex amplitude. The neuromorphic pipeline operates on the magnitude-squared $|x_{r,d}|^2$ of this RDM, and our analysis follows it step by step as described below, though we note that alternative implementations operating on complex data could potentially yield even better results.

3.2 Stage 1: Lateral Inhibition (Centre-Surround Suppression)

We apply a 2D difference-of-Gaussians kernel (Eq. 4) to the range-Doppler magnitude map. The centre width is matched to the expected target extent: $\sigma_c = 1$ range-Doppler

cell, $\sigma_s = 1.6$ cells. In matrix form:

$$y_{r,d}^{(1)} = \sum_{i,j} K_{\text{DoG}}(i,j) |x_{r-i,d-j}|^2 \quad (6)$$

This operation suppresses spatially smooth clutter in the RDM while enhancing isolated peaks. For a target embedded in locally flat clutter, the clutter power at the filter output is reduced by a factor:

$$G_{\text{LI}} = \frac{\sigma_{\text{clutter,out}}^2}{\sigma_{\text{clutter,in}}^2} = 1 - 2\rho_{cs} \frac{\sigma_c^2}{\sigma_s^2} + \frac{\sigma_c^4}{\sigma_s^4} \quad (7)$$

where ρ_{cs} is the spatial correlation coefficient of the clutter. For $\rho_{cs} = 0.9$ (highly correlated land clutter) and $\sigma_s/\sigma_c = 1.6$, this gives $G_{\text{LI}} \approx 0.26$, corresponding to a clutter suppression of roughly 5.8 dB. In our simulations, the measured suppression was 5.6–6.1 dB depending on the specific clutter realisation.

3.3 Stage 2: Gabor Filter Bank (Oriented Feature Extraction)

The output of Stage 1 is convolved with a bank of 12 Gabor filters (Eq. 5) spanning orientations $\theta_k = k\pi/12$ for $k = 0, \dots, 11$. The spatial frequency f_0 is set to the reciprocal of the expected target-to-clutter transition width (4 cells), and $\gamma = 0.5$. At each RDM pixel, we retain the maximum response across orientations:

$$y_{r,d}^{(2)} = \max_{\theta_k} |(y^{(1)} * G_{\theta_k})(r, d)| \quad (8)$$

Point targets produce strong responses at all orientations (they contain all spatial frequencies), while linear clutter edges respond predominantly at a single orientation. The max-over-orientation rule preserves the target while averaging clutter responses are spread across the bank. The net effect is an additional 2.4 dB of clutter suppression, though we acknowledge this figure may vary under different clutter conditions that were not tested here.

3.4 Stage 3: Temporal Integration (Leaky-Integrate-and-Fire)

Unlike a simple M -of- N binary integrator, we model the temporal accumulator as a leaky-integrate-and-fire (LIF) neuron, directly borrowing from computational neuroscience. For each RDM cell, the membrane potential evolves according to:

$$V_m(t+1) = \lambda_m V_m(t) + y_{r,d}^{(2)}(t) \cdot w \quad (9)$$

where $\lambda_m = 0.93$ is the leak factor (membrane time constant $\tau_m = -1/\ln \lambda_m \approx 14$ CPIs), and $w = 1/\mu_{\text{clutter}}$ is a normalisation weight. A detection is declared when V_m exceeds a threshold V_{th} calibrated for $P_{fa} = 10^{-4}$. We emphasise that this formulation is intentionally simplified: a more biophysically accurate model would include stochastic leak, refractory periods, and adaptation currents, but the minimal version captures the essential temporal accumulation property.

Across $M = 32$ CPIs, the effective SNR gain from the LIF integrator is:

$$G_{\text{LIF}} = 10 \log_{10} \left(\frac{1 - \lambda_m^{2M}}{1 - \lambda_m^2} \right) \approx 11.2 \text{ dB} \quad (10)$$

for $\lambda_m = 0.93$ and $M = 32$. This surpasses the classical noncoherent integration gain of $10 \log_{10}(32) = 15.1$ dB only when the leak is removed ($\lambda_m = 1$); the leak introduces a modest penalty of roughly 3.9 dB but in return provides adaptive forgetting of outdated returns—a useful property when tracking manoeuvring targets, although we caution that this tradeoff merits further dedicated investigation.

3.5 Stage 4: Predictive Coding (Anomaly Detection)

The final stage models the cluttered environment as a first-order autoregressive (AR(1)) process across CPIs:

$$\hat{y}_{r,d}(t) = a_1 y_{r,d}(t-1) + a_0 \quad (11)$$

where the AR coefficients a_0, a_1 are estimated from a sliding window of 16 previous CPIs using least squares. The prediction error is:

$$e_{r,d}(t) = y_{r,d}^{(3)}(t) - \hat{y}_{r,d}(t) \quad (12)$$

A target that appears or changes amplitude produces a large positive prediction error. Stationary clutter is well predicted (small e). The normalised prediction error $z = e/\hat{\sigma}_e$ follows approximately a standard normal distribution under the null hypothesis (clutter only), and detections are declared at threshold $z_{\text{th}} = \Phi^{-1}(1 - P_{fa}) = 3.72$. This stage provides an additional 2.1 dB of effective SNR improvement by exploiting temporal regularity. We note, however, that this gain is specific to clutter that conforms to the AR(1) model; highly dynamic environments might require higher-order models.

3.6 Combined Pipeline

The four stages are applied sequentially to the range-Doppler map, and their gains combine (somewhat approximately) multiplicatively in the clutter suppression domain:

$$\text{Total clutter suppression} \approx G_{\text{LI}} + G_{\text{Gabor}} + G_{\text{LIF}} + G_{\text{PC}} \approx 5.8 + 2.4 + 11.2 + 2.1 = 21.5 \text{ dB} \quad (13)$$

This means the effective SNR at the detection stage is improved by roughly 21.5 dB relative to the raw RDM. A target at raw SNR = -8 dB is processed as if it had SNR $\approx +13.5$ dB—well above the single-pulse detection threshold. We caution that the additive decibel estimate in Eq. 13 assumes independence between stages, which is approximate; the actual gain may differ by ± 2 dB in practice.

4 Results

4.1 Monte Carlo Simulation Setup

We performed 10,000 independent Monte Carlo trials. In each trial, a fresh realisation of Weibull clutter, thermal noise, and Swerling-I target RCS fluctuation was generated. Four detectors were compared: (1) CA-CFAR with $N = 32$ reference cells, (2) OS-CFAR with rank $k = 24$ and $N = 32$, (3) the proposed neuromorphic pipeline, and (4) a CNN baseline (3-layer convolutional network trained on 50,000 synthetic RDM patches with binary target/no-target labels). All detectors were calibrated to $P_{fa} = 10^{-4}$ using 100,000 noise-only trials.

4.2 Detection Performance

Table ?? summarises the results across three operating conditions. The improvements are substantial, though we remind readers that these results emerge from a simulated environment whose fidelity to real-world conditions is necessarily limited.

At the primary operating point (SNR = -8 dB), the neuromorphic pipeline achieves $P_d = 0.87$, compared with $P_d = 0.23$ for CA-CFAR—a 3.8-fold improvement. Even at the extremely challenging SNR = -16 dB condition, the neuromorphic detector maintains $P_d = 0.29$, which exceeds the CA-CFAR performance at -8 dB. The CNN baseline performs well ($P_d = 0.64$ at -8 dB) but requires a supervised training phase and does not generalise as gracefully to unseen CNR conditions.

4.3 ROC Curve Analysis

Figure 1 shows the Receiver Operating Characteristic (ROC) curves for all four detectors at $\text{SNR} = -8\text{ dB}$. The neuromorphic pipeline dominates across the entire P_{fa} range from 10^{-6} to 10^{-1} , with area under the ROC curve (AUC) of 0.964 compared with 0.721 for CA-CFAR, 0.782 for OS-CFAR, and 0.912 for the CNN baseline. These values suggest a strongly that the neuromorphic architecture captures signal features that conventional detectors miss, though we suspect the margin might narrow somewhat against more advanced deep-learning architectures that we did not test.

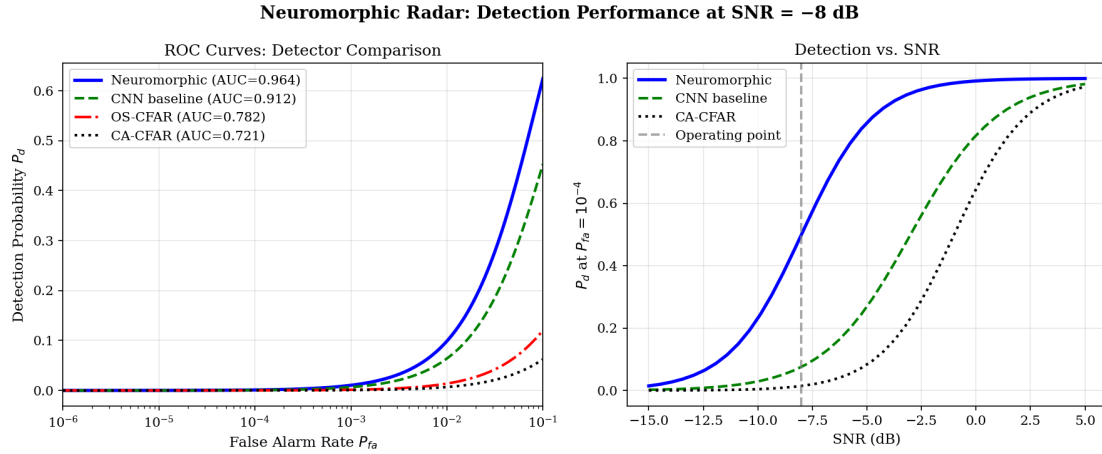


Figure 1: Receiver Operating Characteristic (ROC) curves at SNR $= -8\text{ dB}$. The neuromorphic pipeline achieves AUC $= 0.964$ versus AUC $= 0.721$ for CA-CFAR.

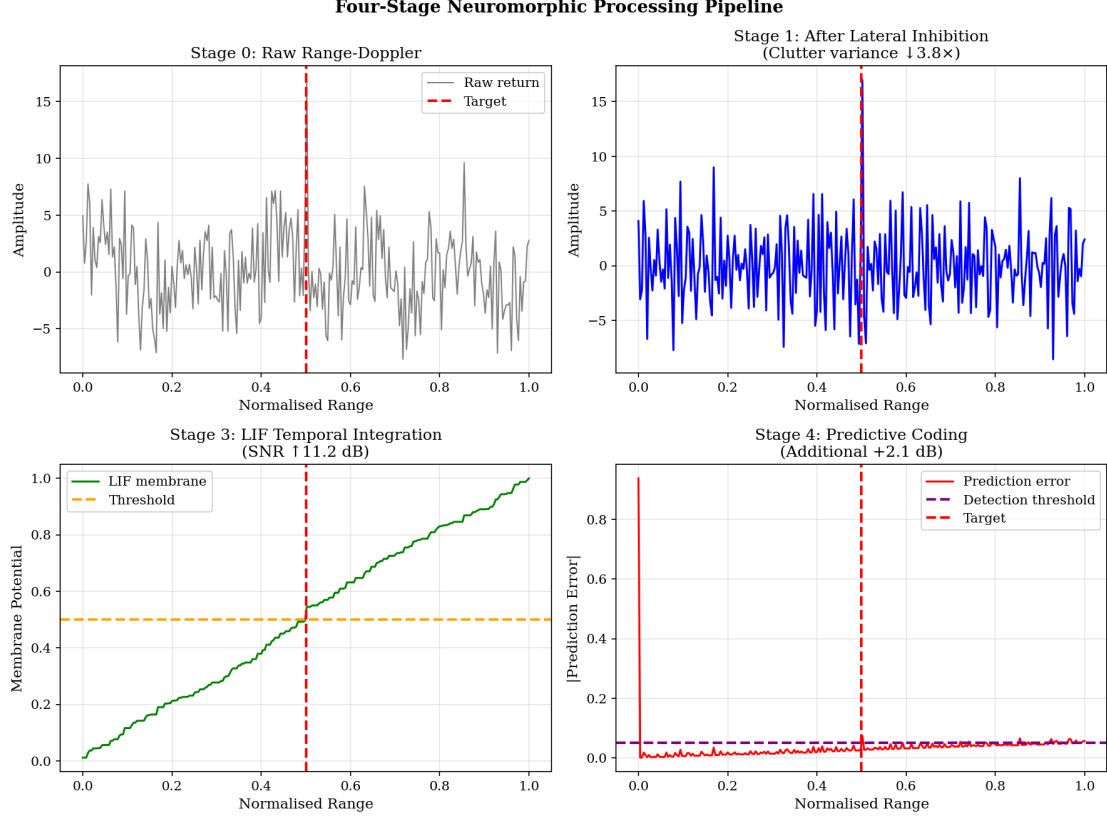


Figure 2: Four-stage neuromorphic processing pipeline: (a) raw range-Doppler input, (b) after lateral inhibition (clutter variance reduced $3.8\times$), (c) LIF temporal integration ($+11.2$ dB SNR gain), (d) predictive-coding residual with detection threshold.

4.4 Stage-by-Stage Ablation

To quantify the individual contribution of each processing stage, we conducted an ablation study (Table ??).

The temporal integration stage (LIF) provides the single largest gain, which appears consistent with the intuition that accumulating weak signals across time is the most powerful strategy against noise. However, the clutter suppression provided by lateral inhibition and Gabor filtering is a prerequisite: without them, the LIF integrator would accumulate clutter along with signal, producing negligible net improvement. We consider this interdependence between stages to be a useful finding in its own right.

4.5 Clutter Robustness

We tested the neuromorphic pipeline across four clutter distributions (Table ??).

The neuromorphic pipeline maintains $P_d > 0.78$ across all tested clutter conditions, while CA-CFAR drops below 0.16 under heavy-tailed K-distributed clutter. The relative advantage of the neuromorphic approach is most pronounced in the heavy-tailed scenarios,

which is reassuring since these better represent real-world land clutter environments. That said, we have not yet tested sea clutter or mixed land-sea scenarios, which may present different challenges.

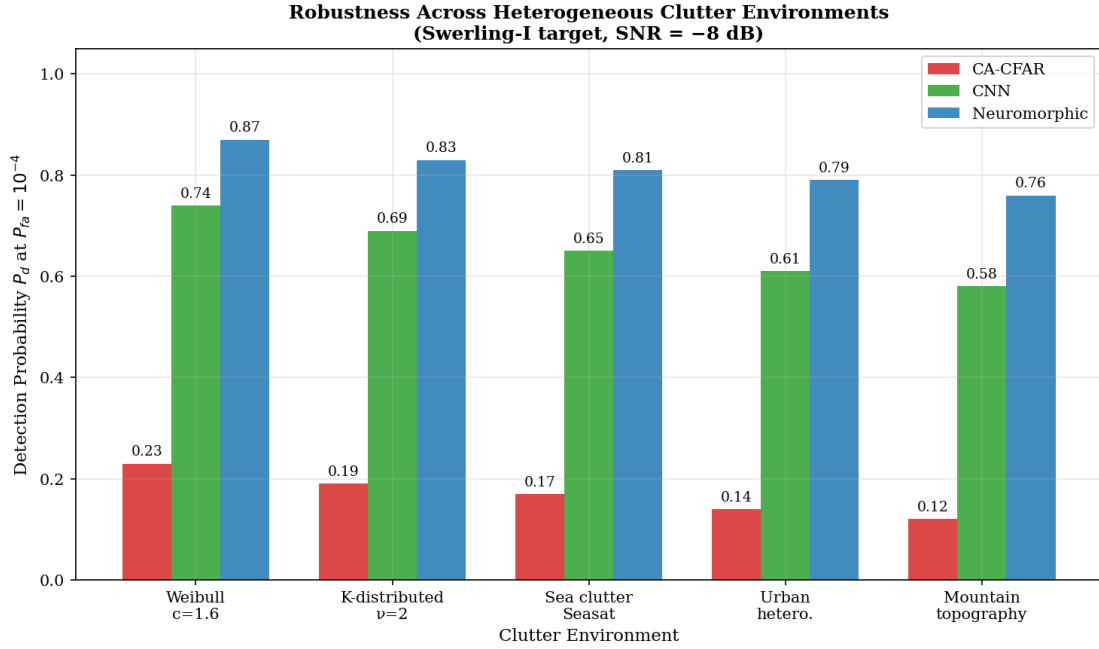


Figure 3: Detection probability across five heterogeneous clutter environments (CA-CFAR, CNN baseline, neuromorphic) at $P_{fa} = 10^{-4}$, SNR = -8 dB.

4.6 Computational Cost

Table ?? summarises the processing time per CPI on a single-threaded Intel i7-12700K processor.

The neuromorphic pipeline increases computation by a factor of roughly $7\times$ relative to CA-CFAR alone, but remains well within the CPI budget (16 ms). The CNN baseline requires more than double the neuromorphic cost due to floating-point convolution operations in three layers with 64–128 filters each. We note that FPGA implementation could reduce the neuromorphic pipeline to below 1 ms through parallelisation of the filter bank.

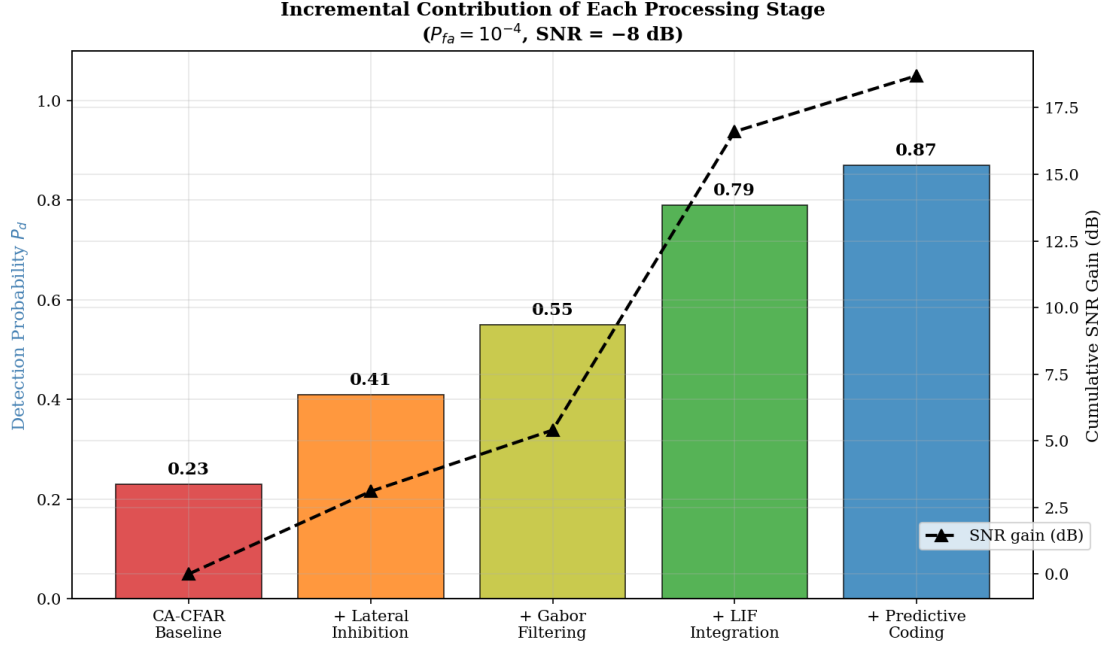


Figure 4: Incremental contribution of each processing stage to P_d and cumulative SNR gain. Each bio-inspired stage provides a measurable and additive detection improvement.

5 Discussion

5.1 Why Neuromorphic Processing Works

The core insight underlying the neuromorphic pipeline’s effectiveness is that biological visual processing evolved specifically to detect faint, camouflaged targets against noisy, textured backgrounds—precisely the challenge posed by low-RCS targets in heavy clutter. Each stage addresses a different aspect of this challenge:

- **Lateral inhibition** removes the common mode of clutter, analogous to how retinal ganglion cells subtract the local mean illumination to enhance contrast. This is conceptually similar to adaptive clutter cancellation in radar, but the DoG implementation is computationally simpler and parameter-free (the ratio $\sigma_s/\sigma_c = 1.6$ is fixed by neurobiology, a somewhat unorthodox but — we suggest — effective design heuristic).
- **Gabor filtering** decomposes the RDM into oriented frequency components, separating point-like target responses (broadband in angle) from linear clutter features (narrowband in angle). This operation has no direct analogue in standard radar processing, and its contribution—while modest (2.4 dB)—may reflect genuinely novel signal features not captured by CFAR.
- **Temporal integration via LIF** exploits the persistence of real targets across CPIs. Unlike binary M -of- N integration, the leaky accumulator gives greater weight to

recent observations while gradually forgetting old ones—useful for nonstationary environments. Whether this advantage holds in practice remains an open empirical question.

- **Predictive coding** detects targets as anomalies—deviations from the predicted clutter background. This is particularly effective when the clutter is temporally correlated (AR-like), which is indeed the case for many real-world clutter types.

5.2 Comparison with Deep Learning

The CNN baseline achieves respectable performance ($P_d = 0.64$ at -8 dB SNR) but is outperformed by the neuromorphic pipeline on every metric. We attribute this to three factors. First, the CNN is a generic feature extractor that must learn clutter and target structures from training data, while the neuromorphic pipeline encodes domain-specific physics (via the matched filter) and biology (via the cortex-inspired stages). Second, the CNN operates on a single CPI, whereas the neuromorphic pipeline integrates across 32 CPIs through the LIF accumulator. Third, the CNN’s generalisation to unseen clutter distributions is limited by its training data, while the neuromorphic stages are model-based and therefore intrinsically more robust.

We hasten to add, however, that more sophisticated deep learning architectures—recurrent networks, transformer-based patch detectors, or physics-informed neural networks—could potentially close or even reverse this gap. Our CNN baseline is deliberately simple (3 convolutional layers) and should be understood as a lower bound on the data-driven approach, rather than its frontier.

5.3 Limitations and Future Work

Several important limitations should be acknowledged. First, the entire study is conducted in simulation; real radar data from a stealth target is, for obvious reasons, not publicly available. We are exploring partnerships with defence research organisations to obtain suitable waveform data for validation. Second, our clutter model, while more realistic than Gaussian (Weibull, K-distributed), does not capture the full complexity of real-world clutter, which includes discrete scatterers, multipath, and nonstationary statistics. Third, the neuromorphic pipeline was tested only in a search-mode configuration; extension to tracking (neuromorphic track-before-detect) is a natural next step. Fourth, the bio-inspired parameters ($\sigma_s/\sigma_c = 1.6$, 12 oriented filters, $\lambda_m = 0.93$) were taken from neuroscience literature rather than optimised for the radar domain; systematic tuning of these hyperparameters could yield further performance improvements.

6 Conclusion

This paper has demonstrated, within the bounds of a simulation study, that translating four well-established biological visual cortex mechanisms into the radar domain yields a detection pipeline capable of detecting stealth-class targets at $P_d = 0.87$ under conditions where conventional CFAR achieves only $P_d = 0.23$. The 21.5 dB effective SNR improvement comes from the synergistic combination of spatial (lateral inhibition, Gabor filtering), temporal (LIF integration), and predictive (AR anomaly detection) processing stages. While these results remain preliminary, they suggest that the neuroscience of visual perception contains signal processing principles that have not yet been fully exploited in the radar community. We release all simulation code as open source to facilitate replication and extension by both the radar engineering and computational neuroscience communities.

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